Homework 1 Theory

Tuesday, August 27, 2024 9:19 PM

1. Often in binary classification we are interested in the differences in the output of our current classifier, g, and an unknown function f that we are trying to learn. It is common in these cases to examine the quantity produced by $f(x)g(x)$ for a given input x. For this problem, let D be an arbitrary distribution on the domain $\{-1,1\}^n$, and let $f, g : \{-1,1\}^n \to \{-1,1\}$ be two Boolean functions.

   (a) **[6 points]** Prove that
   $$\mathbb{P}_{x\sim D}[f(x) \neq g(x)] = \frac{1 - \mathbb{E}_{x\sim D}[f(x)g(x)]}{2}.$$

   (b) **[4 points]** Would this still be true if the domain were some other domain (such as $\mathbb{R}^n$, where $\mathbb{R}$ denotes the real numbers, with say the Gaussian distribution) instead of $\{-1,1\}^n$? If yes, justify your answer. If not, give a counterexample.
   Note: Only the *domain* changes here. The output is still boolean.

1. (a) $\mathbb{P}_{x\sim D}[f(x) \neq g(x)] = \frac{1-\mathbb{E}_{x\sim D}[f(x)g(x)]}{2}$

$f, g : \{-1,1\}^n \to \{-1,1\}$

	$f(x)g(x)$
$f(x) = g(x)$	1
$f(x) \neq g(x)$	-1

$\mathbb{E}_{x\sim D}[f(x)g(x)] = 1 \cdot \mathbb{P}_{x\sim D}[f(x)=g(x)] + (-1)\mathbb{P}_{x\sim D}[f(x)\neq g(x)]$ ⟨def. of Expectation⟩
$= (1 - \mathbb{P}_{x\sim D}[f(x)\neq g(x)]) - \mathbb{P}_{x\sim D}[f(x)\neq g(x)]$ ⟨Probability of complements⟩
$= 1 - 2\,\mathbb{P}_{x\sim D}[f(x)\neq g(x)]$ ⟨algebra⟩

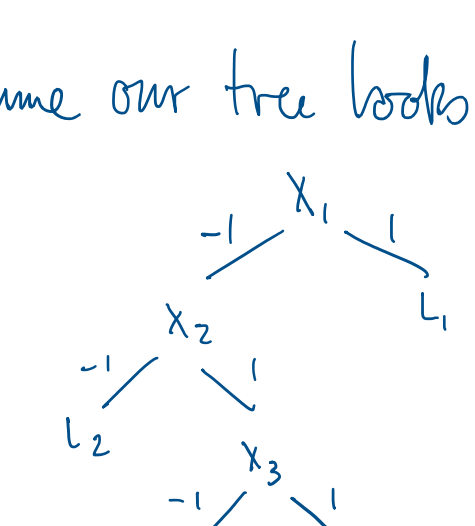
$\Leftrightarrow \mathbb{P}_{x\sim D}[f(x)\neq g(x)] = \frac{1-\mathbb{E}_{x\sim D}[f(x)g(x)]}{2}$ ⟨algebra⟩

(b) Yes, the proof is agnostic to the domain.
It only depends on the boolean output.
No matter the support of X, $f(x)=g(x) \Rightarrow f(x)g(x) = 1$
and $f(x)\neq g(x) \Rightarrow f(x)g(x) = -1$

2. **[10 points]** Let f be a decision tree with t leaves over the variables $x = (x_1, \ldots, x_n) \in \{-1,1\}^n$. Explain how to write f as a multivariate polynomial $p(x_1, \ldots, x_n)$ such that for every input $x \in \{-1,1\}^n$, $f(x) = p(x)$. (You may interpret -1 as FALSE and 1 as TRUE or the other way round, at your preference.) *(Hint: try to come up with an "indicator polynomial" for every leaf, i.e. one that evaluates to the leaf's value if x is such that that path is taken, and 0 otherwise.)*

Let $X = \{X_1, X_2, X_3\}$

Assume our tree looks like this:



then polynomial representation would be:

$p(X) = L_1 \cdot \mathbb{1}[X_1 = 1]$ (left)
$+ L_2 \cdot \mathbb{1}[X_1 = -1] \cdot \mathbb{1}[X_2 = -1]$ (left, left)
$+ L_3 \cdot \mathbb{1}[X_1 = -1] \cdot \mathbb{1}[X_2 = 1] \cdot \mathbb{1}[X_3 = -1]$ (left, right, left)
$+ L_4 \cdot \mathbb{1}[X_1 = -1] \cdot \mathbb{1}[X_2 = 1] \cdot \mathbb{1}[X_3 = 1]$ (left, right, right)

We can do this generally for tree $f(x)$ as defined.

Let L_i, $i \in [1, 2, \ldots, t]$ = value of leaf i
Define an indicator I_i for each L_i indicating the path as above
Then, $p(X) = \sum_{i=1}^{t} L_i I_i$

3. **[10 points]** Compute a depth-two decision tree for the training data in table 1 using the Gini function, $C(a) = 2a(1-a)$ as described in class. What is the overall accuracy on the training data of the tree? For clarity, this will be a full binary tree and a full binary tree of depth-two has four leaves.

X	Y	Z	Number of positive examples	Number of negative examples
0	0	0	10	20
0	0	1	25	5
0	1	0	35	15
0	1	1	35	5
1	0	0	5	15
1	0	1	30	10
1	1	0	10	10
1	1	1	15	5

Table 1: decision tree training data

1. Root node:

base case: $C(\Pr[\text{NEG}]) = 2(85/165)(1-85/165) = 0.4995$ total: 250 NEG: 85

X: $\Pr[X=0]\,C(\Pr_{X=0}[\text{NEG}]) + \Pr[X=1]\,C(\Pr_{X=1}[\text{NEG}])$
$= (150/250)\,C(45/150) + (100/250)\,C(40/100)$
$= 0.252 + 0.192 = 0.4440$
$X=0$: 150, $X=1$: 100; $\text{NEG}_{X=0}$: 45, $\text{NEG}_{X=1}$: 40

Y: $\Pr[Y=0]\,C(\Pr_{Y=0}[\text{NEG}]) + \Pr[Y=1]\,C(\Pr_{Y=1}[\text{NEG}])$
$= (120/250)\,C(50/120) + (130/250)\,C(35/130)$
$= 0.2333 + 0.2046 = 0.4379$
$Y=0$: 120, $Y=1$: 130; $\text{NEG}_{Y=0}$: 50, $\text{NEG}_{Y=1}$: 35

Z: $\Pr[Z=0]\,C(\Pr_{Z=0}[\text{NEG}]) + \Pr[Z=1]\,C(\Pr_{Z=1}[\text{NEG}])$
$= (120/250)\,C(60/120) + (130/250)\,C(25/130)$
$= 0.24 + 0.1615 = 0.4015$
$Z=0$: 120, $Z=1$: 130; $\text{NEG}_{Z=0}$: 60, $\text{NEG}_{Z=1}$: 25

Gain(Z) = 0.4995 − 0.4015 = 0.0980 → split on Z

2. Left node (Z=0):

base case: $C(\Pr_{Z=0}[\text{NEG}]) = 2\cdot(60/120)\cdot(1-60/120) = 0.5000$ $Z=0$: 120 $\text{NEG}_{Z=0}$: 60

X: $\Pr_{Z=0}[X=0]\,C(\Pr_{X=0,Z=0}[\text{NEG}]) + \Pr_{Z=0}[X=1]\,C(\Pr_{X=1,Z=0}[\text{NEG}])$
$= (80/120)\,C(35/80) + (40/120)\,C(25/40)$
$= 0.3281 + 0.1563 = 0.4844$
$X=0, Z=0$: 80, $X=1, Z=0$: 40; $\text{NEG}_{X=0,Z=0}$: 35, $\text{NEG}_{X=1,Z=0}$: 25

Y: $\Pr_{Z=0}[Y=0]\,C(\Pr_{Y=0,Z=0}[\text{NEG}]) + \Pr_{Z=0}[Y=1]\,C(\Pr_{Y=1,Z=0}[\text{NEG}])$
$= (50/120)\,C(35/50) + (70/120)\,C(25/70)$
$= 0.175 + 0.2679 = 0.4429$
$Y=0, Z=0$: 50, $Y=1, Z=0$: 70; $\text{NEG}_{Y=0,Z=0}$: 35, $\text{NEG}_{Y=1,Z=0}$: 25

Gain(Y) = 0.5000 − 0.4429 = 0.0571 → split on Y

right node (Z=1):

base case: $C(\Pr_{Z=1}[\text{NEG}]) = 2(25/130)(1-25/130) = 0.3107$ $Z=1$: 130 $\text{NEG}_{Z=1}$: 25

X: $\Pr_{Z=1}[X=0]\,C(\Pr_{X=0,Z=1}[\text{NEG}]) + \Pr_{Z=1}[X=1]\,C(\Pr_{Y=1,Z=1}[\text{NEG}])$
$= (70/130)\,C(10/70) + (60/130)\,C(15/60)$
$= 0.1319 + 0.1731 = 0.3049$
$X=0, Z=1$: 70, $X=1, Z=1$: 60; $\text{NEG}_{X=0,Z=1}$: 10, $\text{NEG}_{X=1,Z=1}$: 15

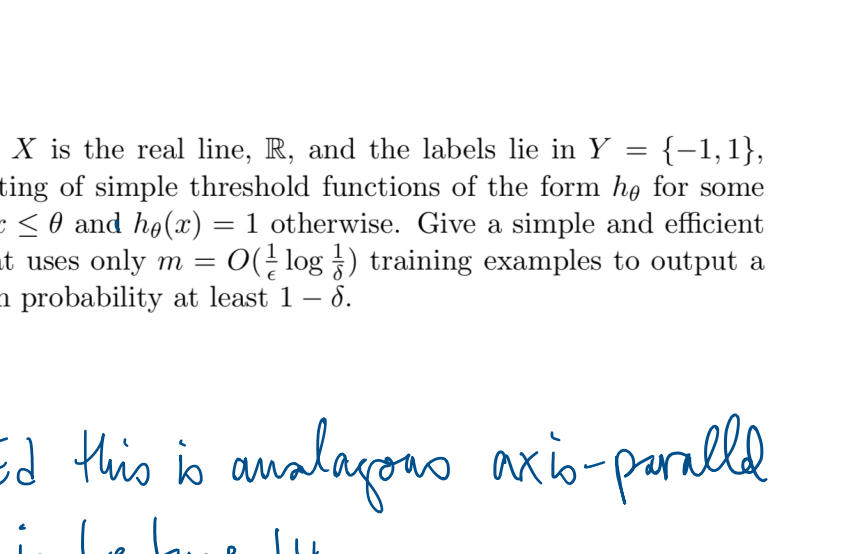
Y: $\Pr_{Z=1}[Y=0]\,C(\Pr_{Y=0,Z=1}[\text{NEG}]) + \Pr_{Z=1}[Y=1]\,C(\Pr_{Y=1,Z=1}[\text{NEG}])$
$= (70/130)\,C(15/70) + (60/130)\,C(10/60)$
$= 0.0659 + 0.1282 = 0.1941$
$Y=0, Z=1$: 70, $Y=1, Z=1$: 60; $\text{NEG}_{Y=0,Z=1}$: 15, $\text{NEG}_{Y=1,Z=1}$: 10

Gain(Y) = 0.3107 − 0.1941 = 0.1166 → split on Y

3. Determine leaf values:

$\text{majority}_{Z=0,Y=0}$(labels) = 0 (35 > 15)
$\text{majority}_{Z=0,Y=1}$(labels) = 1 (45 > 25)
$\text{majority}_{Z=1,Y=0}$(labels) = 1 (55 > 15)
$\text{majority}_{Z=1,Y=1}$(labels) = 1 (50 > 10)

4. Accuracy



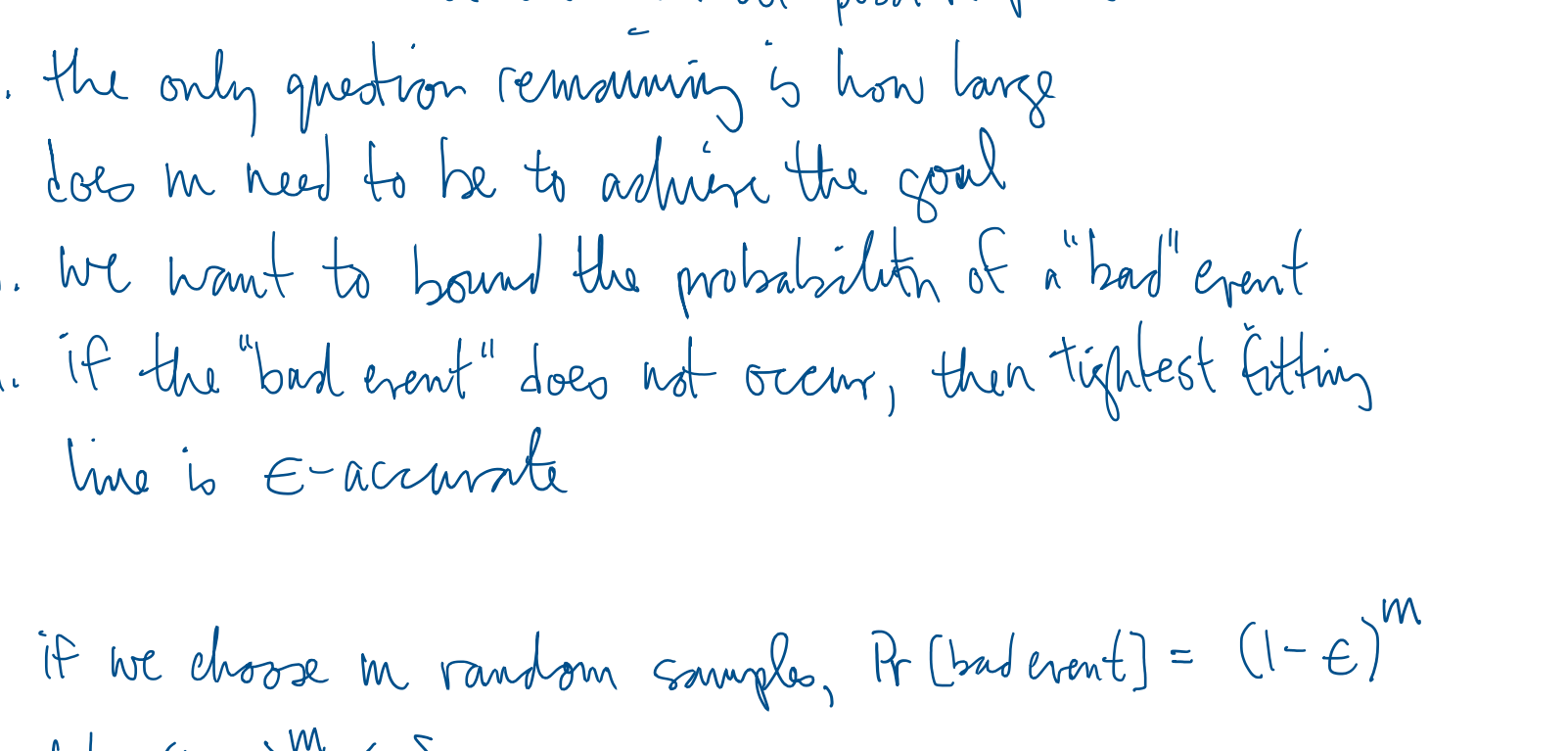
correct: 35 + 45 + 55 + 50 = 185

accuracy: 185/250 = 0.74

4. **[10 points]** Suppose the domain X is the real line, $\mathbb{R}$, and the labels lie in $Y = \{-1,1\}$. Let $\mathcal{C}$ be the concept class consisting of simple threshold functions of the form h_θ for some $\theta \in \mathbb{R}$, where $h_\theta(x) = -1$ for all $x \le \theta$ and $h_\theta(x) = 1$ otherwise. Give a simple and efficient PAC learning algorithm for $\mathcal{C}$ that uses only $m = O(\frac{1}{\epsilon}\log\frac{1}{\delta})$ training examples to output a classifier with error at most ϵ with probability at least $1-\delta$.

* staff suggested on Ed this is analogous axis-parallel rectangles problem in Lecture 1.4

goal: given ϵ, δ, output h that is ϵ-accurate with prob $\ge 1-\delta$



1. choose the θ nearest the left most positive point
2. the only question remaining is how large does m need to be to achieve the goal
3. we want to bound the probability of a "bad" event
4. if the "bad event" does not occur, then tightest fitting line is ϵ-accurate

if we choose m random samples, Pr[bad event] = $(1-\epsilon)^m$

let $(1-\epsilon)^m \le \delta$ $\quad 1+x \approx e^x$, $1-x \approx e^{-x}$

$e^{-\epsilon m} \le \delta$

$-\epsilon m \le \ln(\delta)$

$m \ge \frac{\ln(\delta)}{-\epsilon} = \frac{1}{\epsilon}\ln\left(\frac{1}{\delta}\right)$

⇒ choosing the tightest-fitting line w/ $m \ge O\left(\frac{1}{\epsilon}\ln\left(\frac{1}{\delta}\right)\right)$ samples is ϵ-accurate w/ prob $(1-\delta)$

5. **[6 points]** In this problem we will show that the existence of an efficient mistake-bounded learner for a class $\mathcal{C}$ implies an efficient PAC learner for $\mathcal{C}$.

Concretely, let $\mathcal{C}$ be a function class with domain $X \in \{-1,1\}^n$ and binary labels $Y \in \{-1,1\}$. Assume that $\mathcal{C}$ can be learned by algorithm/learner A with some mistake bound t. You may assume you know the value t. You may also assume that at each iteration, A runs in time polynomial in n and that A only updates its state when it gets an example wrong. The **concrete goal of this problem** is to create a PAC-learning algorithm, B, that can PAC-learn concept class $\mathcal{C}$ with respect to an arbitrary distribution D over $\{-1,1\}^n$ using algorithm A as a sub-routine.

In order to prove that learner B can PAC-learn concept class $\mathcal{C}$, we must show that there exists a finite number of examples, m, that we can draw from D such that B produces a hypothesis whose true error is more than ϵ with probability at most δ. First, fix some distribution D on X, and we will assume that the examples are labeled by an unknown $c \in \mathcal{C}$. Additionally, for a hypothesis (i.e. function) $h : X \to Y$, let $\text{err}(h) = \mathbb{P}_{x\sim D}[h(x) \neq c(x)]$. Formally, we will need to bound m such that the following condition holds:

$$\forall \delta, \epsilon \in [0,1], \exists m \in \mathbb{N} \mid \mathbb{P}_{x\sim D}[\text{err}(B(\{x\}^m)) > \epsilon] \le \delta \qquad x \sim D \tag{1}$$

where $B(\{x\}^m)$ denotes a hypotheses produced from B with m random draws of x from an arbitrary distribution D.

To find this m, we will first decompose it into blocks of examples of size k and make use of results based on a single block to find the bound necessary for m that satisfies condition 1.

Note: Using the identity $\mathbb{P}[\text{err}(h) > \epsilon] + \mathbb{P}[\text{err}(h) \le \epsilon] = 1$, we can see that $\mathbb{P}[\text{err}(h) > \epsilon] \le \delta \Leftrightarrow \mathbb{P}[\text{err}(h) \le \epsilon] \ge 1-\delta$, which makes the connection to the definition of PAC-learning discussed in lecture explicit.

(a) Fix a single arbitrary hypothesis $h' : X \to Y$ produced by A and determine a lower bound on the number of examples, k, such that $\mathbb{P}[\text{err}(h') > \epsilon] \le \delta'$. (The contrapositive view would be: with probability at least $1-\delta'$, it must be the case that $\text{err}(h') \le \epsilon$. Make sure this makes sense.)

err(h') is the true error

assume $\text{err}(h') > \epsilon \Rightarrow P(h' \text{ correctly classifies } x) \le (1-\epsilon)$

$\Rightarrow P[h' \text{ and } \text{err}(h') > \epsilon] \le (1-\epsilon)^k$

bound $(1-\epsilon)^k \le \delta'$

$e^{-\epsilon k} \le \delta'$

$-\epsilon k \le \ln(\delta')$

$k \ge \frac{\ln(\frac{1}{\delta'})}{\epsilon}$

to guarantee $\text{err}(h') \le \epsilon$ w/ probability $> 1-\delta'$

(b) From part 5a we know that as long as a block is at least of size k, then if that block is classified correctly by a *fixed arbitrary hypothesis* h' we can effectively upper bound the probility of the 'bad event' (i.e. A outputs h' s.t. $\text{err}(h') > \epsilon$) by δ'. However, our bound must apply to every h that our algorithm B could output for an arbitrary distribution D over examples. With this in mind, how large should m be so that we can bound all hypotheses that could be output? (You may assume that algorithm B will know the mistake bound throughout the question.)

- A makes at most t mistakes
- only updates when it makes a mistake
- A will return at most t+1 hypotheses
- need $m = (t+1)k$
  $= (t+1)\frac{\ln(\frac{1}{\delta'})}{\epsilon}$ samples

(c) Put everything together and fully describe (with proof) a PAC learner that is able to output a hypothesis with a true error at most ϵ with probability at least $1-\delta$, given a mistake bounded learner A. To do this you should first describe your pseudocode for algorithm B which will use A as a sub-routine (no need for minute details or code, broad strokes will suffice). Then, prove there exists a finite number of m examples for B to PAC-learn $\mathcal{C}$ for all values of δ and ϵ by lower bounding m by a function of ϵ, δ, and t (i.e. finding a finite lower bound for m such that the PAC-learning requirements in 1 are satisfied).

1 gather sample of size m
2 pass sample of size $k < m$ to A
3 if A succeeds on k examples, we're done because it has achieved its objective (true error can only be $> \epsilon$ with prob $\le (1-\delta)$ – to be proven)
4 else go back to 2 with the next k samples
5 ends if we make t mistakes

Proof:

1 $k \ge \frac{\ln(\frac{1}{\delta'})}{\epsilon} \Rightarrow \text{err}(h') > \epsilon$ w/ prob. $\le \delta'$ (part a)

2 let $(t+1)\delta' = \delta$
$\delta' = \frac{\delta}{t+1}$

3 $\Rightarrow k \ge \frac{\ln(\frac{t+1}{\delta})}{\epsilon}$

4 let $m = (t+1)\frac{\ln(\frac{t+1}{\delta})}{\epsilon}$ samples

5 if we pass $k = \frac{\ln(\frac{t+1}{\delta})}{\epsilon}$ samples into A and are correct, this only would happen w/ prob $\frac{\delta}{t+1}$

6 Prob.(5) happens w/ one of the t+1 hypotheses is $(t+1)\left(\frac{\delta}{t+1}\right) = \delta$